

Contact Dynamics Hybrid Systems Trajectory Optimization

Sanghyun Kim

Kyung Hee University
RCI Lab

- LQR with Quaternions
- Tangent space reduction
- $SE(3)$ control

Contact Dynamics Problem

In flight:

$$m\ddot{z} = -mg$$

Smooth ODE.

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Velocity discontinuity.

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→ Cannot model impact with a smooth ODE.

Two Approaches

1) Hybrid (Event-based)

- Integrate ODE
- Detect guard function $g(x) = 0$
- Apply jump map

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2) Time-stepping (Contact-implicit)

- No event detection
- Solve constrained optimization each step
- Contact force computed explicitly

Backward Euler Discretization

$$m \frac{v_{k+1} - v_k}{h} = -mg + J^T \lambda$$

$$q_{k+1} = q_k + hv_{k+1}$$

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Contact constraints:

$$\phi(q_{k+1}) \geq 0$$

$$\lambda \geq 0$$

$$\lambda \phi(q_{k+1}) = 0$$

Complementarity Condition

$$\lambda \geq 0$$

$$\phi(q) \geq 0$$

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Meaning:

- No penetration
- No pulling contact
- Force only when in contact

Time-stepping $\simeq$ KKT conditions of:

$$\min_{v_{k+1}, \lambda} \frac{1}{2} m \|v_{k+1} - v_k\|^2$$

subject to contact constraints.

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Widely used in:

- PyBullet
- MuJoCo
- Gazebo

Hybrid Model Structure

Continuous dynamics:

$$\dot{x} = f(x)$$

Guard function:

$$g(x) = 0$$

Jump map:

$$x^+ = \Delta(x^-)$$

Falling Brick Example

Guard:

$$z = 0$$

Jump map:

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Advantages:

- Exact impact time
- High accuracy integrators
- Clean for trajopt

Hybrid TrajOpt Formulation

Divide trajectory into modes:

Mode 1: flight

$$x_{k+1} = f_f(x_k, u_k)$$

Mode 2: stance

$$x_{k+1} = f_s(x_k, u_k)$$

At transition:

$$x^+ = \Delta(x^-)$$

If Mode Sequence Known

If contact times known:

- No guard function needed
- Optimize smooth dynamics only
- Enforce jump constraints explicitly

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Very common in legged locomotion.

Time-Stepping vs Hybrid

	Time-Stepping	Hybrid
Impact time accuracy	Low	Exact
Contact force	Explicit	Implicit
Optimization difficulty	Hard	Easier
Mode sequence needed	No	Yes

Contact introduces:

- Non-smooth dynamics
- Complementarity constraints
- Mode switching

Hybrid modeling makes system piecewise smooth.

Applications:

- Legged locomotion
- MPC with contact
- DIRCOL
- Hybrid LQR

Summary

- Contact cannot be modeled with smooth ODE
- Two major approaches exist
- Hybrid methods dominate trajectory optimization
- Complementarity central to contact modeling